

QUIZ 3 A

Max Marks: $10 \times 1.5 = 15$

Name: _____

Average: 1.97 / 15

ID: _____

S.No.	Question	Answer
1	Consider a technology in which the length of the transistor can vary by a maximum of 18 % from nominal. (That means the length can increase or decrease by this amount). Assuming that only the variation is length is dominant and assuming a long channel model, find the ratio of the minimum current to the maximum current that can flow through this transistor. Given $k'_n = 200 \mu A/V^2$, $V_{Tn} = 0.4V$ and $V_{DD} = 1.8V$.	0.695
2	Consider an NMOS transistor with $W = 1.8 \mu m$ and $L = 0.6 \mu m$ operating with $V_{DS} = 30mV$. We want to use this as a resistor to generate a resistance of $10 k\Omega$. Find the gate voltage in V that needs to be applied given $V_{Tn} = 0.4V$ and $k'_n = 200 \mu A/V^2$. Assume the square law model. $V_S = 0V$	0.5667V
3	Assume that you have a capacitor that is being charged with a constant dc current source of $3.5 \mu A$. The capacitance is voltage dependent given by $C = \alpha V^{-0.5}$ where $\alpha = 0.9 pFV^{0.5}$ and V is the voltage across the capacitor in volts. Find the time in ns, for the capacitor to charge from 0 to 0.5V.	363.65ns
4	Assume that you are discharging a load capacitance of $0.4 pF$ charged up to $V_{DD} = 1.8V$ through an NMOS whose $L = 0.5 \mu m$. Find the width of the transistor needed in μm , if I need to ensure that 6% of the voltage is discharged in 15ps. The technology parameters are $k'_n = 120 \mu A/V^2$ and $V_T = 0.5V$.	28.402
5	Assume that an inverter has been designed for matched rising and falling delays. If the PMOS and NMOS both have a maximum variation of 12 % in their widths from nominal and if the mobilities of both the holes and electrons can vary as much as 15 % from the nominal values, what is the worst-case propagation delay you can expect? Express this delay as a multiple of the nominal delay. You can assume that the variation in W and μ are independent of each other.	1.3369
6	The smallest inverter in a given CMOS technology has $W_p/W_n = 2/2$. What is the logical effort of a 4- input NAND gate. (Enter your answer as a decimal and not as a fraction)	2.5
7	A 51-stage ring oscillator has a time period of 4.08 ns and a duty cycle of 50.123%. What is the τ_{PHL} of this technology in ps?	45ps
8	A 101 stage RO utilizing the smallest inverters in the technology has a delay of 4.5 ns and a duty cycle of 50%. An enthusiastic student decides to use the smallest 2-input NAND gate with one of the inputs held at '1' instead of the inverters in the ring oscillator. What is the delay of this new NAND2 RO in ns? (Assume nominal delays for the stacked transistors and ignore the intermediate node cap in the stack) The smallest inverter for matched pull down and pull up resistance is sized as $W_p/W_n = 2/1$.	7.5 ns
9	What is the min delay (in F04 delays) required to drive a wire load of 1pF with an inverter whose input capacitance is 12fF? Assume that W_p/W_n of the min sized inverter is 3/1 for matched rise and fall resistances.	3.2171
10	What is the output resistance (in kOhms) of the smallest 2-input NAND gate when both inputs are at '0'? $V_{DD} = 1.8V$, $V_{Tp} = -0.4V$, $k'_n = 3k'_p = 180 \mu A/V^2$. The smallest inverter is sized for equal rise and fall delays.	1.9841k Ω

QUIZ 3 B

Max Marks: $10 \times 1.5 = 15$

QUIZ 3 B

Max Marks: $10 \times 1.5 = 15$

Name: _____

ID: _____

S.No.	Question	Answer
1	Consider a technology in which the length of the transistor can vary by a maximum of 15 % from nominal. (That means the length can increase or decrease by this amount). Assuming that only the variation is length is dominant and assuming a long channel model, find the ratio of the minimum current to the maximum current that can flow through this transistor. Given $k'_n = 180 \mu A/V^2$, $V_{Tn} = 0.4V$ and $V_{DD} = 1.8V$.	0.739
2	Consider an NMOS transistor with $W = 1.5 \mu m$ and $L = 0.4 \mu m$ operating with $V_{DS} = 50mV$. We want to use this as a resistor to generate a resistance of $10 k\Omega$. Find the gate voltage in V that needs to be applied given $V_{Tn} = 0.4V$ and $k'_n = 200 \mu A/V^2$. Assume the square law model. $V_S = 0V$	0.5333V
3	Assume that you have a capacitor that is being charged with a constant dc current source of $5 \mu A$. The capacitance is voltage dependent given by $C = \alpha V^{-0.5}$ where $\alpha = 0.7 pFV^{0.5}$ and V is the voltage across the capacitor in volts. Find the time in ns , for the capacitor to charge from 0 to 0.6V.	216.89ns
4	Assume that you are discharging a load capacitance of $0.1 pF$ charged up to $V_{DD} = 1.8V$ through an NMOS whose $L = 0.7 \mu m$. Find the width of the transistor needed in μm , if I need to ensure that 6% of the voltage is discharged in 12ps. The technology parameters are $k'_n = 120 \mu A/V^2$ and $V_T = 0.5V$.	8.8757 μm
5	Assume that an inverter has been designed for matched rising and falling delays. If the PMOS and NMOS both have a maximum variation of 10 % in their widths from nominal and if the mobilities of both the holes and electrons can vary as much as 20 % from the nominal values, what is the worst-case propagation delay you can expect? Express this delay as a multiple of the nominal delay. You can assume that the variation in W and μ are independent of each other.	1.3889
6	The smallest inverter in a given CMOS technology has $W_p/W_n = 3/2$. What is the logical effort of a 5- input NAND gate. (Enter your answer as a decimal and not as a fraction)	2.6
7	A 51-stage ring oscillator has a time period of $6.63 ns$ and a duty cycle of 49.774% . What is the τ_{pHL} of this technology in ps ?	50 ps
8	A 101 stage RO utilizing the smallest inverters in the technology has a delay of $5.8 ns$ and a duty cycle of 50%. An enthusiastic student decides to use the smallest 2-input NAND gate with one of the inputs held at '1' instead of the inverters in the ring oscillator. What is the delay of this new NAND2 RO in ns ? (Assume nominal delays for the stacked transistors and ignore the intermediate node cap in the stack) The smallest inverter for matched pull down and pull up resistance is sized as $W_p/W_n = 2/1$.	9.6669ns
9	What is the min delay (in FO4 delays) required to drive a wire load of $1pF$ with an inverter whose input capacitance is $25fF$? Assume that W_p/W_n of the min sized inverter is $3/1$ for matched rise and fall resistances.	3.336
10	What is the output resistance (in kOhms) of the smallest 2-input NAND gate when both inputs are at '0'? $V_{DD} = 1.8V$, $V_{Tp} = -0.4V$, $k'_n = 3k'_p = 300 \mu A/V^2$. The smallest inverter is sized for equal rise and fall delays.	1.1905k Ω

QUIZ 3 C

Max Marks: $10 \times 1.5 = 15$ Name: ID:

S.No.	Question	Answer
1	Consider a technology in which the length of the transistor can vary by a maximum of 22 % from nominal. (That means the length can increase or decrease by this amount). Assuming that only the variation in length is dominant and assuming a long channel model, find the ratio of the minimum current to the maximum current that can flow through this transistor. Given $k'_n = 180 \mu A/V^2$, $V_{Tn} = 0.4V$ and $V_{DD} = 1.8V$.	0.639
2	Consider an NMOS transistor with $W = 1.8 \mu m$ and $L = 0.6 \mu m$ operating with $V_{DS} = 40mV$. We want to use this as a resistor to generate a resistance of $8 k\Omega$. Find the gate voltage in V that needs to be applied given $V_{Tn} = 0.4V$ and $k'_n = 200 \mu A/V^2$. Assume the square law model. $V_S = 0V$	0.6083V
3	Assume that you have a capacitor that is being charged with a constant dc current source of $6 \mu A$. The capacitance is voltage dependent given by $C = \alpha V^{-0.5}$ where $\alpha = 0.3 pFV^{0.5}$ and V is the voltage across the capacitor in volts. Find the time in ns, for the capacitor to charge from 0 to $0.4V$.	63.246ns
4	Assume that you are discharging a load capacitance of $0.1 pF$ charged up to $V_{DD} = 1.8V$ through an NMOS whose $L = 0.3 \mu m$. Find the width of the transistor needed in μm , if I need to ensure that 9% of the voltage is discharged in 10ps. The technology parameters are $k'_n = 140 \mu A/V^2$ and $V_T = 0.5V$.	13.694 μm
5	Assume that an inverter has been designed for matched rising and falling delays. If the PMOS and NMOS both have a maximum variation of 18 % in their widths from nominal and if the mobilities of both the holes and electrons can vary as much as 10 % from the nominal values, what is the worst-case propagation delay you can expect? Express this delay as a multiple of the nominal delay. You can assume that the variation in W and μ are independent of each other.	1.355
6	The smallest inverter in a given CMOS technology has $W_p/W_n = 2/3$. What is the logical effort of a 5- input NAND gate. (Enter your answer as a decimal and not as a fraction)	3.4
7	A 51-stage ring oscillator has a time period of $5.1 ns$ and a duty cycle of 50.196%. What is the τ_{pHL} of this technology in ps?	60ps
8	A 101 stage RO utilizing the smallest inverters in the technology has a delay of 10ns and a duty cycle of 50%. An enthusiastic student decides to use the smallest 2-input NAND gate with one of the inputs held at '1' instead of the inverters in the ring oscillator. What is the delay of this new NAND2 RO in ns? (Assume nominal delays for the stacked transistors and ignore the intermediate node cap in the stack) The smallest inverter for matched pull down and pull up resistance is sized as $W_p/W_n = 2/1$.	16.667ns
9	What is the min delay (in FO4 delays) required to drive a wire load of $1pF$ with an inverter whose input capacitance is $5fF$? Assume that W_p/W_n of the min sized inverter is $3/1$ for matched rise and fall resistances.	3.8085
10	What is the output resistance (in kOhms) of the smallest 2-input NAND gate when both inputs are at '0'? $V_{DD} = 1.8V$, $V_{Tp} = -0.4V$, $k'_n = 3k'_p = 210 \mu A/V^2$. The smallest inverter is sized for equal rise and fall delays.	1.7007k Ω